

Which Latents Are Worth Predicting?

A conceptual manuscript on predictive signal and transfer in autoregressive self-supervision

Abstract

An autoregressive model can learn from a token and from its own representation of that token. The second form of supervision offers a potentially richer training signal, but its usefulness depends on which internal representation is chosen. We formulate target selection as a question about the interaction between predictive structure and learning dynamics. A representation may preserve a relevant latent variable while exposing little of it to prediction; conversely, its prediction loss may improve a model before the relevant variable is decodable from the current state. We develop a framework that separates target content, prediction from a fixed representation, and transfer through shared parameters. For normalized latent regression, we derive an exact decomposition of prediction difficulty, show why fixed-state predictability does not characterize feature learning, and identify a branch-averaging intervention that preserves the expected auxiliary gradient. A finite-sample clustering result formalizes when noisy observations expose hierarchical distinctions, while a local update identity describes the additional conditions required for beneficial optimization. These arguments motivate experiments on ordinary causal Transformers trained on hierarchical synthetic languages, followed by language-model pretraining. The empirical objective is to determine whether a target’s changing predictive signal explains its causal training value and supports useful adaptation under a fixed compute budget.

1. Introduction

Next-token prediction trains a language model to summarize a prefix well enough to predict its continuation. This objective supplies a complete probabilistic training criterion, but it does not uniquely prescribe the representations or computational strategies by which predictions are produced. Under limited data and computation, different objectives can encourage different internal solutions even when they ultimately support the same token distribution. Auxiliary prediction of learned representations is one way to influence this process.

NITP augments token prediction with prediction of a future shallow-layer representation and reports a preference for contextualized early layers in its tested configurations (Zhang et al., 2026). NextLat instead learns token-conditioned transitions between hidden states, motivated by the formation of sufficient summaries of history (Teoh et al., 2026). These approaches make different choices about what is predicted and what information the predictor receives. Their comparison raises a broader question: **what makes one learned representation more useful supervision than another?**

Layer depth is a convenient experimental variable because it changes target representations without requiring a different backbone. It is not itself a definition of abstraction. A shallow layer can acquire increasingly structured features over time; a deeper layer can preserve several scales of information or specialize toward a narrower output computation. Nor does a representation’s ability to encode a latent variable determine the benefit of predicting it. A training signal acts through a particular source representation, predictor, parameterization, optimizer, and stage of learning.

Consider a sequence whose distant continuation depends on a hidden grammatical category, while its immediate next token contains substantial lexical variation. A target could encode the category perfectly but devote most of its normalized geometry to that unpredictable lexical detail. A second target could encode the category less comprehensively yet expose it more strongly to the prediction objective. Which target is a better teacher cannot be resolved by decoding accuracy alone. Even successful prediction does not settle the question: learning a predictable nuisance feature need not improve the language model.

Our thesis is that the usefulness of a latent target is relational. It depends on the predictive distinctions the target exposes and on the parameter updates through which the learner can exploit

them. The proposed project studies this relation using a controlled hierarchical data model and ordinary causal Transformer layers. Known latent variables are used to interpret representations and design diagnostic interventions; the training architecture receives no parse-aligned slots or latent labels.

The resulting research program has two linked objectives. The scientific objective is to explain changes in the causal value of latent supervision. The practical objective is to determine whether these changes can be estimated economically enough to improve training. A curriculum that moves toward deeper layers is one possible outcome. A fixed layer whose representation becomes progressively more useful is another.

This manuscript develops the conceptual framework and proves several elementary statements used to organize the experiments. It does not report completed training experiments. Its proposed empirical contribution is the connection between these statements, identifiable hierarchical structure, and measured transfer in end-to-end Transformers.

2. Problem formulation

Let $C = X_{\leq t}$ denote a random observed prefix and $Y = X_{t+1}$ its next token. A D -block causal Transformer with parameters θ produces a source state

$$S_\theta(C) = h_t^D(C).$$

A target is constructed from block j of a teacher with parameters $\bar{\theta}$ after it has observed the next token:

$$Z_j(C, Y) = \frac{h_{t+1}^{j, \bar{\theta}}(C, Y)}{\|h_{t+1}^{j, \bar{\theta}}(C, Y)\|_2}.$$

For clarity, assume the vectors being normalized are nonzero. Practical implementations use a declared numerical tolerance. The predictor q_{ϕ_j} maps the source state into a normalized prediction

$$U_j(C) = u_{\phi_j}(S_\theta(C)) = \frac{q_{\phi_j}(S_\theta(C))}{\|q_{\phi_j}(S_\theta(C))\|_2}.$$

The auxiliary loss is

$$\mathcal{L}_j(\theta, \phi_j; \bar{\theta}) = \mathbb{E}[1 - U_j(C)^\top Z_j(C, Y)],$$

and the training objective combines it with ordinary autoregressive cross-entropy:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{NTP}} + \sum_j \lambda_j \mathcal{L}_j, \quad \lambda_j \geq 0.$$

The primary configuration uses the current model as teacher and stops gradients through the target. Thus $\bar{\theta}$ is set to the current parameters when a target is constructed, but is treated as constant for that update. All arguments below involving differentiation explicitly hold the teacher fixed. A moving teacher changes subsequent updates, so a fixed-teacher argument is a local statement about the training dynamics, not a proof of convergence for the coupled system. The exact diagnostic also assumes deterministic evaluation of the teacher; additional randomness must be fixed or integrated separately.

To define usefulness, let $\mathcal{S}_s$ contain the complete training state at checkpoint s , including optimizer and predictor states. Let $\mathcal{U}_a^{(k)}$ continue training for k updates under action $a = (j, \lambda)$, and let $\mathcal{U}_0^{(k)}$ use only NTP. For a predeclared outer loss J , define

$$G_a(\mathcal{S}_s; k) = \mathbb{E} \left[J(\mathcal{U}_0^{(k)}(\mathcal{S}_s)) - J(\mathcal{U}_a^{(k)}(\mathcal{S}_s)) \right].$$

The expectation is over a specified continuation protocol. Paired runs use identical sampled mini-batches and common random numbers wherever possible. Positive G_a means that the intervention improves the outer objective relative to the baseline. It is conditional on the checkpoint, continuation horizon, auxiliary strength, and evaluation distribution. It is not an intrinsic property of block j .

For scientific comparisons, k and the sampled tokens are matched. For efficiency comparisons, the total computational budget is matched instead, including all target-selection work. These estimands answer different questions and should both be reported.

3. What can a latent prediction objective learn?

3.1 Prediction from a fixed source state

Fix a checkpoint and suppress the target index. Write $S = S_\theta(C)$, and define two conditional means:

$$m_C(C) = \mathbb{E}[Z \mid C], \quad m_S(S) = \mathbb{E}[Z \mid S].$$

Because S is a function of C , the tower property gives $m_S(S) = \mathbb{E}[m_C(C) \mid S]$. The first mean describes target components that survive averaging over the next-token branches. The second describes what survives the additional compression into the current source state.

Proposition 1 (fixed-state prediction). Over all measurable unit-vector predictors $u(S)$,

$$\inf_u \mathbb{E}[1 - u(S)^\top Z] = 1 - \mathbb{E}\|m_S(S)\|_2.$$

An optimum is $u^*(S) = m_S(S)/\|m_S(S)\|_2$ whenever $m_S(S) \neq 0$; any unit vector is optimal when the mean is zero.

Proof. Conditional expectation gives $\mathbb{E}[u(S)^\top Z] = \mathbb{E}[u(S)^\top m_S(S)]$. Cauchy-Schwarz bounds the inner product by $\|m_S(S)\|_2$, and the stated predictor attains the bound. $\square$

This is a statement about normalized regression. It does not say that predicting the conditional mean models the full distribution of future representations. Related distinctions between expectation prediction and distributional self-prediction arise in the analysis of self-predictive reinforcement learning (Ni et al., 2024).

The proposition yields a more detailed decomposition. Let $\mathcal{Q}$ be a declared family of unit-vector predictors and let $\hat{u} \in \mathcal{Q}$ be the predictor produced by a particular fitting procedure. Define

$$A_{\mathcal{Q}} = \sup_{u \in \mathcal{Q}} \mathbb{E}[u(S)^\top m_S(S)].$$

Then its population loss is exactly

$$\begin{aligned} \mathcal{L}(\hat{u}) = & \underbrace{1 - \mathbb{E}\|m_C\|}_{\text{branch ambiguity}} \\ & + \underbrace{\mathbb{E}\|m_C\| - \mathbb{E}\|m_S\|}_{\text{compression into the source state}} \\ & + \underbrace{\mathbb{E}\|m_S\| - A_{\mathcal{Q}}}_{\text{restriction of the predictor family}} \\ & + \underbrace{A_{\mathcal{Q}} - \mathbb{E}[\hat{u}(S)^\top m_S]}_{\text{gap of the fitted predictor}}. \end{aligned}$$

All terms are nonnegative. The first follows from $\|Z\| = 1$, the second from conditional Jensen, and the last two from the definition of the supremum. The final term includes whatever finite-data and optimization limitations remain after choosing $\mathcal{Q}$; it does not separate those limitations without an additional analysis.

This decomposition motivates distinct experiments. Branch averaging isolates sampling variation associated with the first term without removing the cancellation in its conditional mean. Changing the source or examining its probes addresses compression. A head-capacity sweep addresses the third term. Equal head-fitting budgets and convergence checks address the fourth. None of these terms alone equals causal training value.

Example: perfect decoding with a weak predictive component. Let the prefix determine $A \in \{-1, +1\}$, and let Y be an independent fair sign. Consider the unit target

$$Z_\gamma = (\gamma A, \sqrt{1 - \gamma^2} Y), \quad 0 < \gamma \leq 1.$$

The first coordinate recovers A perfectly for every positive γ . Nevertheless,

$$m_C = (\gamma A, 0), \quad \inf_{u(C)} \mathcal{L} = 1 - \gamma.$$

As γ decreases, target decodability remains perfect while the component exposed to prediction shrinks. This does not prove that the target becomes harmful. It establishes that decodability and predictive signal are different quantities. Whether the shrinking component matters depends on the future task and the learner's update directions.

3.2 A fixed-state probe does not characterize feature learning

The previous argument fixes the source representation. Training the backbone changes that representation. This distinction is essential because two prefixes that currently produce the same hidden state can have different parameter derivatives.

Let $J_\theta(C) = D_\theta U(C)$ be the prediction Jacobian with respect to shared backbone parameters, holding the predictor parameters and teacher fixed. Under differentiability and integrability conditions that permit exchanging expectation and differentiation,

$$\nabla_\theta \mathcal{L} = -\mathbb{E}[J_\theta(C)^\top Z] = -\mathbb{E}[J_\theta(C)^\top m_C(C)].$$

In general, m_C cannot be replaced by m_S in this expression. The Jacobian can depend on features of C that are absent from the value of S . The omitted term would be

$$-\mathbb{E}[J_\theta(C)^\top \{m_C(C) - m_S(S(C))\}],$$

which need not vanish. For a head-only update, by contrast, the head Jacobian is a function of S , so conditioning on S is sufficient.

Example: learning before a variable is decodable. Let $C = A$, where A is a fair sign, and let the source be $S_\theta(A) = \theta A$. Take the frozen target $Z(A) = (A, 0)$ and either predictor

$$u_\sigma(s) = (\sigma \sin s, \cos s), \quad \sigma \in \{-1, +1\}.$$

Both are unit vectors. At $\theta = 0$, every prefix produces $S = 0$. No head can decode A from this state, and $m_S(0) = 0$. Yet

$$\mathcal{L}_\sigma(\theta) = 1 - \sigma \sin \theta, \quad \mathcal{L}'_\sigma(0) = -\sigma.$$

The target supplies a nonzero learning signal even though the relevant variable is not currently decodable from the source.

The same example separates target content from transfer. Suppose the primary task predicts A using θA , with loss $J(\theta) = \frac{1}{2}(\theta - 1)^2$. An auxiliary-only step from the fixed reference point $\theta = 0$ gives $\theta^+ = \eta\lambda\sigma$, so

$$J(0) - J(\theta^+) = \eta\lambda\sigma - \frac{1}{2}\eta^2\lambda^2.$$

For a sufficiently small step, one predictor helps and the other hurts. The target, current state, current predictor outputs, and auxiliary loss values are identical in the two cases. Their parameter derivatives differ. This is a local counterexample, not a claim about their eventual convergence, but it rules out a universal target-only criterion for the sign of immediate transfer.

The conceptual implication is constructive. A useful analysis should examine which target distinctions correlate with the learner’s available update directions. A decodability threshold may explain a particular staged algorithm, but it cannot be assumed to be a prerequisite for useful end-to-end supervision.

3.3 An exact intervention on next-token uncertainty

On a small synthetic grammar, the distribution $p_\star(y \mid C)$ can be computed and all next-token targets can be evaluated. Define

$$\bar{Z}(C) = \sum_y p_\star(y \mid C) Z(C, y) = m_C(C).$$

Proposition 2 (branch-averaged supervision). Replacing the sampled auxiliary loss by $1 - U(C)^\top \bar{Z}(C)$ preserves its conditional expected loss and its conditional expected gradient through the predictor and backbone, with the teacher held fixed.

Proof. Both the loss and its gradient through U are linear in the target. Condition on C and take the expectation over Y . $\square$

The average must remain unnormalized. Normalizing $\bar{Z}$ would reweight prefixes according to its norm and would define a different objective. An implementation must therefore use the displayed dot product, rather than a cosine routine that normalizes both arguments.

There is also an exact variance statement. Let $g(C, Y)$ be the auxiliary gradient and $\bar{g}(C) = \mathbb{E}[g(C, Y) \mid C]$. The law of total covariance gives

$$\text{Cov}(g) = \text{Cov}(\bar{g}) + \mathbb{E}[\text{Cov}(g \mid C)].$$

Thus branch averaging weakly reduces auxiliary gradient covariance in the positive-semidefinite order. It removes sampling variation, while retaining any cancellation already present in the mean target. When the NTP gradient uses the same sampled next token, cross-covariance between the two losses can also change. The covariance of their sum is not guaranteed to decrease. This qualification is experimentally testable using independent conditional samples for the two losses as an additional oracle control.

Providing the realized next token to the predictor is a complementary intervention. It changes the available information and can reveal branch-specific structure, but also changes the task. In particular, determinism of Z given (C, Y) does not imply determinism given (S, Y) . Token-conditioned latent prediction is central to NextLat and has a related uncertainty motivation in EAGLE (Teoh et al., 2026; Li et al., 2024). Branch averaging provides the more controlled test of target-sampling noise because it preserves the auxiliary population semi-gradient.

4. Hierarchical data as a model of predictive structure

4.1 Why use the Random Hierarchy Model?

The Random Hierarchy Model (RHM) generates visible tokens by recursively expanding hidden symbols through a regular tree. Each parent has several permitted child tuples, or synonyms. These alternatives provide a concrete distinction between the identity of an abstract variable and the details of its realization. Prior work connects learning in this model to correlations that expose the shared parent of synonymous tuples (Cagnetta and Wyart, 2024; Cagnetta et al., 2025).

The RHM is useful here because it supplies known structure against which learned targets can be interpreted. It does not require the model to be told that structure. The main experiment uses ordinary causal Transformers and raw leaf sequences. Network layer j , true hierarchy level r , and training checkpoint s remain separate axes.

The grammar also makes controlled interventions possible. A production-rule swap changes a subtree’s visible realization while preserving its parent. A parent-symbol change alters the underlying category. Comparing responses to these interventions distinguishes synonym invariance from general insensitivity. For causal states, the altered subtree must lie in the observed prefix unless future-invariance itself is being checked.

At an arbitrary prefix, some ancestral variables remain uncertain even for an observer who knows the grammar. Evaluation should therefore compare a probe to the posterior $p_\star(H^{(r)} \mid C)$, or evaluate fully observed subtrees at their completion points. Otherwise an information limit can be misidentified as a learning failure.

4.2 A statistical statement with explicit assumptions

The following result formalizes one component of readiness: when observed features expose a partition into synonym classes. It concerns a specified mean-based recovery procedure, not the complete Transformer training dynamics.

Fix a local grammar neighborhood. Let T range over N grammatical source tuples, with parent label $a(T)$. Let Z be a vector-valued observation in another view. Choose a fixed linear transform W , independent of the samples used for estimating means, and write $V = WZ$. Define

$$\nu_t = \mathbb{E}[V \mid T = t].$$

Assume there are parent centers μ_a such that $\|\nu_t - \mu_{a(t)}\|_2 \leq b$, and distinct centers have separation at least $\delta > 0$. The quantity b measures the defect in synonym invariance. Suppose that, for every tuple, the centered observations satisfy

$$\|V - \nu_t\|_2 \leq R \quad \text{almost surely,} \quad \mathbb{E}[\|V - \nu_t\|_2^2 \mid T = t] \leq \sigma^2.$$

For each tuple, collect at least n conditionally independent observations and form its empirical mean $\hat{\nu}_t$. Dependence between the collections for different tuples does not affect the union bound below.

Proposition 3 (finite-sample exposure of a synonym partition). For $0 < \zeta < 1$, define

$$\epsilon_n = \frac{\sigma}{\sqrt{n}} + R\sqrt{\frac{2\log(N/\zeta)}{n}}.$$

If $b + \epsilon_n < \delta/4$, then, with probability at least $1 - \zeta$, the graph connecting empirical means at Euclidean distance less than $\delta/2$ has exactly the true synonym classes as its connected components.

Proof. For one tuple, independence and centering give $\mathbb{E}\|\hat{\nu}_t - \nu_t\|^2 \leq \sigma^2/n$, so Jensen bounds the expected norm by $\sigma/\sqrt{n}$. Replacing one observation changes the norm of the mean error by at most $2R/n$. The bounded-differences inequality therefore bounds its upper-tail deviation by

$R\sqrt{2\log(N/\zeta)/n}$ with failure probability at most ζ/N . A union bound controls every tuple mean simultaneously. Each empirical mean is then within $b + \epsilon_n$ of its parent center. Means in the same class are separated by less than $\delta/2$, whereas means in different classes are separated by more than $\delta/2$. The graph consists of one clique per class. $\square$

This is a sufficient condition for a particular recovery rule. Failure of the inequality does not imply information-theoretic impossibility. The bound is deliberately elementary rather than optimally sharp. Its useful feature is that it tracks separation, within-class structure, and observation fluctuations in the same metric. The variance term is determined by $\text{tr}(W \text{Cov}(Z \mid T = t) W^\top)$; a whitening or projection estimated on an independent split can change which distinctions are emphasized.

In an ideal separated-view RHM construction, synonyms have identical population means. If tuples occur with probability of order $1/(vm)$, the effective count per tuple is of order $P/(vm)$. Combining bounded observation noise with a stipulated separation of order $\sqrt{1-f}/m$, where $f = m/v^{s-1}$ for branching factor s , yields a sufficient scale

$$P \gtrsim \frac{vm^3}{1-f} \log \frac{vm}{\zeta},$$

in addition to a sample-coverage requirement. This recovers the structure of the local clustering scale used by [Korchinski, Favero, and Wyart \(2026\)](#). Their hierarchical recovery argument has further assumptions and scope restrictions; importing this local scale does not establish the same sample complexity for adjacent causal latent prediction.

4.3 Why a confusion matrix is not enough

Suppose a categorical external variable Y is observed through a channel Q that is conditionally independent of the source tuple given Y . Then its transformed context means factor through Q . If Q contracts a relevant parent distinction by a factor α while leaving observation noise of constant order, an inverse- α^2 sufficient sample bound can follow from Proposition 3. The noise qualification is essential.

For example, replace a one-hot observation by

$$Z_c = c e_Y + (1 - c)\mathbf{1}/v, \quad c > 0.$$

This deterministic transformation shrinks every centered mean difference, R , and σ by the same factor c . The sufficient condition in Proposition 3 is unchanged. The observation is also invertible for every positive c . A hard corruption channel that randomly emits the wrong symbol can instead destroy information. Identical-looking mean contractions therefore need not imply identical learning thresholds.

A learned causal target may violate the channel factorization because it also encodes the source tuple and surrounding prefix. In that case the invariance defect b is itself part of what must be measured. If both views are independently corrupted, their cross-covariance takes a two-sided form $R_0 K Q^\top$, rather than depending on a target channel alone. Here R_0 denotes the source observation channel, distinct from the radius R in Proposition 3.

The proposed statistical work is to identify relevant modes, quantify their separation and noise under controlled corruptions, and test their recovery. The further research problem is to show how these modes contribute to the actual auxiliary gradients of a causal learner. Proposition 3 does not assume that detecting a correlation guarantees that gradient descent will extract it.

5. From predictive signal to beneficial updates

Let $\bar{g}_j$ be the population auxiliary gradient on shared parameters at a fixed checkpoint. Section 3 shows that it is determined by the interaction of the target’s full-prefix conditional mean and the prediction Jacobian. To characterize its local effect, consider a fixed baseline parameter vector θ_0^+ and an additional update $-\eta\lambda\hat{d}_j$. Let $\hat{d}_j$ have mean d_j and covariance V_j/n . The direction can incorporate a fixed preconditioner; the baseline and preconditioner are conditioned on throughout this statement.

Proposition 4 (local transfer). If the outer loss J is quadratic with Hessian H , then

$$\begin{aligned} \mathbb{E}[J(\theta_0^+) - J(\theta_0^+ - \eta\lambda\hat{d}_j)] &= \eta\lambda \nabla J(\theta_0^+)^\top d_j \\ &\quad - \frac{\eta^2\lambda^2}{2} \left(d_j^\top H d_j + \frac{\text{tr}(H V_j)}{n} \right). \end{aligned}$$

Proof. Expand the quadratic exactly around θ_0^+ and use $\mathbb{E}[\hat{d}_j^\top H \hat{d}_j] = d_j^\top H d_j + \text{tr}(H V_j)/n$. $\square$

For a smooth nonlinear loss whose Hessian is locally L_H -Lipschitz, the absolute error in this expression is at most

$$\frac{L_H \eta^3 \lambda^3}{6} \mathbb{E} \|\hat{d}_j\|_2^3,$$

provided the bound holds along every relevant update segment and the third moment is finite. This specifies when the quadratic description is informative rather than treating it as a global theory.

The first term measures alignment with the outer objective. The second incorporates deterministic curvature and stochastic update effects. When H is positive semidefinite, the latter are penalties. In a nonconvex region, they need not be. Under positive curvature, a target can have useful mean signal but still be harmful at excessive weight or high variance. Conversely, immediate alignment can be weak while a longer sequence of updates changes the representation in a beneficial way.

The proposition also motivates choosing auxiliary strength. Write $A = \nabla J(\theta_0^+)^\top d_j$ and $B = d_j^\top H d_j + \text{tr}(H V_j)/n$. When $B > 0$, the maximizer of the quadratic gain over $0 \leq \lambda \leq \lambda_{\max}$ is

$$\lambda^* = \min \left\{ \lambda_{\max}, \max \left\{ 0, \frac{A}{\eta B} \right\} \right\}.$$

This is a local model, not a proposed universal weight formula. Exact Adam updates alter moments and clipping behavior, and must be evaluated using their actual state transitions. The identity explains why both target choice and weight belong in the intervention definition.

Gradient-based selection and validation lookahead have established precedents (Du et al., 2018; Liu et al., 2022). The new scientific objective is to connect the mean and covariance terms to identifiable predictive distinctions, then verify the resulting predictions under interventions. A generic alignment statistic with a new name would not establish that connection.

6. Hypotheses and experimental consequences

The framework leads to several testable hypotheses without requiring a monotone depth curriculum.

Predictive-mode hypothesis. Target value changes when an intervention alters a task-relevant component of m_C or its coupling to update directions. Removing an equally sized but irrelevant component should have a smaller effect, after controlling changes in scale and predictor conditioning. This comparison requires matched interventions and independent identification of the relevant component.

Noise hypothesis. Some differences between targets reflect the covariance of auxiliary updates. Exact branch averaging preserves the auxiliary mean update and changes its variance, providing a

controlled test. The prediction concerns variation and transfer, not a guarantee that every averaged target improves the full training objective.

Accessibility hypothesis. Changing predictor capacity or fitting budget can change target rankings with the source and target representations frozen. Whether this also changes backbone transfer is a separate measured outcome. A predictor that fits a target more accurately is not necessarily a better auxiliary teacher.

Developmental hypothesis. Target value changes with training stage because the target, source representation, and parameter derivatives evolve. A frontier in network depth is one possible pattern. A broad band, a nonmonotone pattern, or a fixed useful layer is consistent with other realizations of the framework. The evidence must determine which occurs.

The primary experiment estimates $G_{j,\lambda}(\mathcal{S}_s; k)$ from paired checkpoint continuations and overlays it with measurements of true hierarchical structure. Interventions then change branch sampling, predictor capacity, target drift, or specific target distinctions while holding the rest of the setup as constant as the question permits. A successful explanation must predict changes on held-out grammars, model seeds, and intervention settings, rather than only fit the original depth sweep.

The principal behavioral endpoint is held-out autoregressive NLL, supplemented on synthetic data by excess loss over the grammar’s conditional entropy. Fixed-prefix future probes predict X_{t+k} from h_t without access to intermediate tokens. They measure a different capability from teacher-forced losses evaluated at later positions. Such probes must have a declared function class, fitting budget, and independent test split; their interpretation is closer to computationally restricted predictive information than to new Shannon information in the prefix (Xu et al., 2020).

If target rankings vary reproducibly, an inexpensive controller can be assessed. It periodically scores candidate actual optimizer updates on an independent selection batch and compares these scores with longer continuations. The relevant test is whether the selected action loses little value relative to the best tested action, and ultimately improves complete trajectories at matched compute. Its selection data and discarded updates count as part of training.

7. Relation to prior work and scope of the contribution

The empirical setting connects several existing lines of work. RHM analyses provide a controlled language model with interpretable hierarchy. The latent-prediction analysis of Korchinski et al. supplies a statistical motivation, while its data2vec experiments concern a bidirectional masked-prediction encoder. Ordinary causal adjacent prediction requires its own empirical and theoretical bridge. The present project does not identify neural depth with grammar depth or assume that a hierarchy-independent clustering bound transfers unchanged to a Transformer.

NITP and NextLat motivate distinct auxiliary objectives. The former provides a direct target-layer question. The latter’s exact consistency conditions explain why token-conditioned transitions can support sufficient representations, but do not settle finite-time target selection. LayerLock already advances latent targets through progressive freezing in vision (Erdogan et al., 2025). HiLP introduces higher-level latent supervision for language rollouts (Shi et al., 2026); only its abstract claim was available for the present comparison, so detailed overlap must be checked before a novelty claim is finalized.

Learning-progress curricula are also established (Graves et al., 2017). They supply useful baselines, but progress on an auxiliary loss is not the same quantity as its causal value to the primary task. Likewise, the four propositions here are analytical foundations, not claims of novel general probability or optimization results. Their role is to make the project falsifiable and to prevent statistical recovery, fixed-state prediction, and end-to-end transfer from being conflated.

The intended contribution is a mechanism supported by causal measurements: an account of which predictive distinctions make a learned target useful, when those distinctions become available to a learner’s updates, and whether this account enables economical target selection. If the adaptive method fails while the mechanism predicts interventions, a substantive scientific result remains

possible. If the mechanism fails, an attractive depth heatmap alone does not establish a readiness theory.

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Implementation and Experimental Plan

Companion to Which Latents Are Worth Predicting?

Objective and experimental logic

The implementation should establish a causal phenomenon, identify the predictive distinctions responsible for it, and only then test whether adaptation improves complete training runs. The conceptual draft separates three objects: information in the target, prediction from a frozen source, and transfer through shared parameters. The experiments should preserve that separation rather than reducing every diagnostic to one readiness score.

The minimum successful scientific result is a reproducible target-value difference that can be predicted under controlled interventions. The stronger practical result is an adaptive rule that improves held-out performance at matched total compute. Neither requires the optimal layer to move monotonically with training.

The first phase uses a standard causal Transformer on raw Random Hierarchy Model (RHM) leaf sequences. Hierarchy variables, exact grammar conditionals, and structural interventions are diagnostic resources. They do not enter the ordinary training loss, architecture, or eventual language-model controller. No large language-model training should begin until the synthetic results justify the expense.

Milestone 0: Fix the estimands and verify the analytical examples

Agree on a small set of outcomes before inspecting target-depth results. The primary outer loss is held-out autoregressive NLL. On synthetic data, also report excess NLL over the grammar’s true conditional entropy, fixed-prefix future prediction, and recovery of causally observable hierarchy variables. These are separate outcomes, not a weighted composite score.

Use the continuation value $G_{j,\lambda}(\mathcal{S}_s; k)$ defined in the conceptual draft. Initially compare matched tokens and updates to identify what an auxiliary target changes. Later compare matched total FLOPs to establish efficiency. Record the checkpoint’s training history, head preparation, target update policy, auxiliary strength, and evaluation distribution as part of every intervention.

Implement small numerical checks of the analytical statements before introducing a Transformer. Verify the fixed-state cosine optimum by enumeration, the example in which opposite predictor Jacobians yield opposite transfer, and the equality of branch-sampled and branch-averaged expected semi-gradients. Check covariance reduction for the auxiliary gradient alone. These checks validate the measurement code and algebra; they are not evidence of a neural-learning phenomenon.

Deliverable: a short experiment specification and executable analytical checks, with the core metrics and data uses fixed.

Completion criterion: each reported metric has an unambiguous conditioning set, continuation horizon, and interpretation. In particular, a fixed-prefix future score must not receive intermediate true tokens.

Milestone 1: Build an observable causal learning system

Implement an RHM generator that returns raw token sequences and a separate diagnostic record containing grammar identity, production choices, latent symbols, and subtree boundaries. Retain the sampled grammar and seeds so that a run can be reconstructed. Use independent grammar draws as experimental replicates.

The generator should support exact inference of $p_\star(y \mid C)$ and relevant latent posteriors for the small configurations. Message passing on the fixed grammar tree is a natural implementation; compare it with exhaustive enumeration on the smallest instances. Do not obtain conditional continuations by blindly resampling a full sequence, which becomes inefficient for unlikely prefixes.

Train a decoder-only Transformer with ordinary positional information. Expose hidden states through a consistent block-index convention. Use a learned predictor from the final state at t to a detached target at $t + 1$. Keep token NTP active. Record both the exact reference cosine formulation and any numerical normalization tolerance. Fit all candidate heads with the same architecture unless head capacity is the intervention.

An initial development configuration is:

Component	Starting choice	Reason
Grammar	Binary tree, depth 5	Several hierarchy levels with short sequences
Vocabulary and rules	$v = 16, m = 4$	Small enough for exact conditional diagnostics
Sequence length	32 leaves	Consequence of the grammar configuration
Transformer	8 blocks, width 256	A modest model with several distinct target depths
Target blocks	$\{0, 1, 2, 4, 6, 8\}$	Embedding, shallow, middle, and final-layer controls
Objective	NTP plus one cosine latent loss	Direct connection to the conceptual analysis

These are development defaults, not known optimal values. First establish stable NTP learning and measurable recovery of more than the shallowest structure on one development grammar. Adjust the development configuration if necessary, then freeze it before confirmation on independent grammars. Do not change the recipe to manufacture a moving optimum.

Use two complementary representation diagnostics. At completed subtrees, fit probes for the corresponding latent identity. For unresolved ancestors, report posterior cross-entropy or Brier excess relative to the grammar posterior. Add synonym-preserving and parent-changing input interventions. Verify that changing only future tokens leaves a causal hidden state unchanged.

Use a fixed finite dataset at several training ages to separate optimization progress from exposure to new samples. Subsequently vary the number of unique sequences under the same training protocol. If heads or an online controller use additional unique sequences, those sequences count toward the method’s learning-data budget.

Deliverable: a reproducible causal training baseline, exact small-grammar inference, and layer-by-hierarchy representation diagnostics.

Completion criterion: stable training, correct causal indexing, validated grammar inference, and nontrivial hierarchical learning. If these are absent, do not begin the large target sweep.

Milestone 2: Measure the causal value of ordinary target layers

Create paired continuation runs from complete checkpoints. A checkpoint must include backbone and predictor parameters, optimizer moments and counters, learning-rate schedule, target-network state if any, data-sampler state, and relevant random-number states. Repeating the same continuation twice is a necessary reproducibility check before interpreting small differences between arms.

Use two checkpoint families: NTP-only training and training with a fixed shallow auxiliary target from initialization. In the second family, include continuing the current target as an explicit arm. The value of switching to j relative to continuing the current target is obtained by comparing those paired arms, rather than treating removal of all auxiliary supervision as the operational baseline.

Head preparation needs its own control. In the main diagnostic, fit candidate heads for an equal budget on cached, detached source and target states. Include the active head in the same declared

calibration protocol. On a subset, repeat with inherited head states or cold starts to assess dependence on head history. A target should not lose a depth comparison merely because its head has not been trained.

At five checkpoints, continue an NTP-only arm and six target arms for 128 updates, recording outcomes at updates 32 and 128. On a subset, add a longer continuation to check whether early rankings persist. Begin with one development-selected auxiliary weight; expand to a small weight sweep where rankings are sensitive. Also compare a subset at matched shared-parameter auxiliary gradient norm. This latter intervention answers a scale-control question and should be labeled separately from the fixed-weight recipe.

An illustrative confirmation campaign uses three grammar draws and two model seeds. For one checkpoint family and one auxiliary weight, the branch budget is

$$3 \times 2 \times 5 \times 7 \times 128 = 26,880$$

shadow updates. The second checkpoint family doubles this count. Base training, head fitting, target evaluation, and uncertainty diagnostics add further cost. Profile one run to turn these counts into a resource estimate; do not describe this campaign as a few-percent overhead.

Report the full value map, uncertainty, and head-preparation conditions. Bootstrap or otherwise aggregate at the independent grammar level with model seeds nested within grammars. Three grammars provide a pilot uncertainty estimate; increase the number before making a broad statistical claim. Evaluate representational changes as outcomes of the intervention, alongside NLL changes.

Deliverable: paired target-value maps across depth, checkpoint, continuation horizon, and training history.

Completion criterion: at least some reproducible benefits or harms larger than the measurement uncertainty and robust to basic head and scale controls. A moving ridge is not required.

Milestone 3: Identify the mechanism through targeted interventions

Select informative checkpoints and target pairs from the development maps, then confirm intervention predictions on held-out grammars. Avoid a full Cartesian product of every ablation. Each intervention should resolve a specific competing explanation.

Mechanism	Main intervention	Measurements and interpretation
Branch sampling	Compute $m_j(C) = \sum_y p_*(y C) Z_j(C, y)$ and use $1 - U_j(C)^\top m_j(C)$	Check equal expected semi-gradients; measure auxiliary covariance and continuation value. Never normalize the mean target.
Information revealed by the next token	Compare unconditioned, token-conditioned, and token-only heads with matched budgets	Distinguish a richer conditional task from a token shortcut. This is not the same intervention as branch averaging.
Fixed-source accessibility	Fit small and larger heads on the same frozen features	Compare population-proxy losses, then separately compare their effects when backbone updates resume.

Mechanism	Main intervention	Measurements and interpretation
Target drift	Compare a fixed checkpoint teacher with moving stop-gradient targets; test EMA only if needed	Separate current content from temporal changes. Match comparisons within each teacher policy and count its cost.
Specific predictive distinctions	Apply controlled channels or remove an identified target subspace	Compare with a matched irrelevant corruption or subspace; preserve dimension and variance as closely as possible.
Pairing and persistence	Shuffle target pairings, compare same-position targets, and measure a simple persistence predictor	Determine whether correct temporal structure explains the effect beyond easy copying or marginal geometry.

For exact branch averaging, use a deterministic teacher and enumerate next tokens only in the synthetic setting. Count the extra target evaluations. If the interpretation depends on total gradient variance, use a small oracle control with independent conditional token samples for NTP and the auxiliary loss; averaging only the auxiliary loss need not reduce the covariance of their sum.

For channel experiments, explicitly distinguish stochastic categorical corruption from deterministic softening. Verify the conceptual draft’s scale-invariance example before interpreting inverse-gain-squared plots. Construct paired channels with similar overall decoding accuracy but different effects on a known parent distinction. These are privileged diagnostic interventions, not candidate deployed methods.

For subspace experiments, identify projections on one diagnostic split and evaluate interventions on another. Match removed dimension and variance, apply the same re-normalization policy, and report resulting gradient norms. A projection can erase correlated features as well as its intended variable, so corroborate important conclusions with a second intervention family.

Test Proposition 3 on its actual mean-clustering task: estimate class separation, observation fluctuations, and within-synonym deviations, then measure partition recovery. Keep this plot separate from the Transformer-value plot. A successful recovery bound does not establish transfer unless the next experiment links the corresponding modes to helpful parameter updates.

Deliverable: an intervention matrix with predeclared predictions and a small set of mechanistic plots connecting target content, gradient behavior, and actual transfer.

Completion criterion: at least one mechanism predicts out-of-setting changes better than target depth or training age alone. Otherwise retain a descriptive empirical result and revise the explanatory claim.

Milestone 4: Validate an estimator of finite-horizon value

Compare inexpensive estimates with the paired continuation values from Milestone 2. The main practical estimator uses the actual optimizer, including clipping and moment updates. From a common state, compute a baseline update θ_0^+ and candidate update θ_a^+ , then evaluate

$$\widehat{G}_a^{\text{step}} = J_{\text{select}}(\theta_0^+) - J_{\text{select}}(\theta_a^+).$$

A cheaper approximation uses their actual parameter difference:

$$\widehat{G}_a^{\text{lin}} = -\nabla J_{\text{select}}(\theta_0^+)^{\top}(\theta_a^+ - \theta_0^+).$$

Also evaluate auxiliary learning progress and raw gradient alignment as baselines. Fixed-prefix future probes are optional explanatory features, not automatically controller inputs. Any fitted combination of diagnostics must be trained and tested on disjoint grammar and checkpoint families. Measure rank agreement with 32-step and 128-step values, detection of harmful targets, and selection regret

$$\text{Regret}(\hat{a}) = \max_{a \in \mathcal{A}} G_a - G_{\hat{a}}.$$

The maximum is over the tested action set and includes NTP-only continuation. It is not a claim about a globally optimal curriculum. Estimate selection and reference values on independent evaluation samples to avoid choosing and judging an action on the same noise. Use longer continuations on a subset to quantify how far the estimator’s useful horizon extends.

Profile the actual cost. Separate the cost of candidate backward passes, selection forward passes, head calibration, and state copying. Select a controller interval M that meets a declared overhead allowance, for example 5%, if the profiler shows that this is feasible. Do not assume that a gradient dot product is cheap merely because its final output is one scalar.

Deliverable: an estimator validation report with regret, horizon dependence, generalization across settings, and profiled cost.

Completion criterion: decisions remain useful at a meaningful continuation horizon and their expected benefit can plausibly exceed selection cost. If only expensive long forks work, keep them as diagnostics rather than claiming an efficient controller.

Milestone 5: Test adaptation on complete trajectories

Implement a conservative controller that periodically calibrates or refreshes candidate heads, tests a small set of (j, λ) actions, and continues with the selected action for M updates. Include $\lambda = 0$. Confirm close decisions on a fresh selection batch, or retain the current action when the measured difference is inconclusive. Restore all shadow optimizer and teacher state before continuing the real trajectory.

Initially choose one target rather than a mixture. If mixtures are later introduced, test joint interventions explicitly because single-target gains do not add. Do not restrict candidates to neighboring layers until the empirical ordering supports that restriction. Compare with a simple fixed shallow target, a tuned best fixed target, a uniform mixture at the same total auxiliary budget, a predetermined shallow-to-deep schedule, and a standard gradient or validation-based weighting method.

Allocate comparable tuning resources to the adaptive and fixed methods. Report both operational training compute and the diagnostic or tuning compute needed to develop the method. For the operational comparison, the NTP baseline receives additional ordinary training within the same total FLOP budget. Report token-matched curves as a separate measure of sample efficiency.

Controller-selection and confirmation data are learning resources. Keep them distinct from development evaluations and an untouched final test set, and include their unique examples in the sample budget. The full adaptive path must be run: a schedule assembled from values measured only along an NTP path is not a valid substitute.

Add a compositional test only with a defined distribution shift. Hold out combinations of production choices while retaining exposure to each constituent rule, verify coverage and marginal frequencies, and evaluate on those held-out combinations. New IID strings from the same grammar are a useful test set, but do not alone establish out-of-distribution composition.

Deliverable: complete learning curves at matched total compute, with strong fixed and mixture baselines and independent seeds.

Completion criterion: a supported efficiency or quality gain over the tuned fixed alternatives. If adaptation fails, omit the algorithmic claim while retaining any validated mechanism.

Milestone 6: Test transfer to language modeling

Use the same ordinary-layer objective and validated estimator without grammar labels, parse boundaries, or exact conditional distributions. Begin with a modest 100-300M-parameter pretraining pilot to establish measurable effects. Profile the token budget and confirm a successful finding at a second depth or scale before claiming a general target-depth principle.

Evaluate held-out NLL versus total FLOPs and tokens. Add fixed-prefix future probes and a small set of downstream tasks with meaningful signal at the selected model size. Retain the same inference backbone if inference neutrality is part of the claim. Before launching the experiment, specify which synthetic mechanism is expected to transfer and what observation would contradict that prediction.

Compare with the best fixed targets and controller baselines from Milestone 5. Include a matched NextLat implementation when comparing token-conditioned and unconditioned objectives, acknowledging the differences in its loss components and head. Complete the full-text comparison with HiLP before finalizing novelty claims; include it if hierarchical rollouts become central. Large-scale reproduction of every neighboring method is not required for a paper whose main claim is a controlled account of ordinary-layer target value.

Deliverable: a language-model confirmation of the proposed mechanism or estimator, with complete compute accounting.

Completion criterion: evidence of transfer beyond the synthetic grammar. Failure limits the scope of the paper; it should not be hidden by expanding the benchmark list.

Milestone dependencies and paper outputs

Milestone	Depends on	Principal paper output
0. Definitions and analytical checks	None	Precise claims and validated measurement identities
1. Observable causal baseline	0	Emergent hierarchy measured across ordinary layers
2. Paired target-value maps	1	Causal effect of target choice across training states
3. Mechanistic interventions	2	Predictive distinctions and update effects that explain value
4. Estimator validation	2, informed by 3	Held-out selection regret and cost
5. Full adaptive trajectories	4	Practical benefit at matched compute, if present
6. Language-model transfer	Credible 3; successful 4-5 for the controller claim	Generality beyond the synthetic setting

The core theory work should progress alongside Milestones 2 and 3. Sharpen the noisy-view bound only when the measured observation family warrants it, and derive how particular grammar modes couple to the learner’s gradients. Avoid spending the theoretical budget proving a sharper partition-recovery theorem whose premises do not describe the causal objective being trained.